

The Triadic Theory of Existence: The Electron Universe, the Loan Against Entropy, and the Philosophy of a Quantized Reality

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Abstract

The Triadic Redox Theory (TRT) unifies chemistry, biology, electromagnetism, thermodynamics, and quantum mechanics under one idea: charge flows from H-like donors to He-like acceptors through Middle Fluids, whenever $\chi_H < \Phi < \chi_{He}$. This paper examines both the power and the philosophy of that unification. We first survey the **electron universe**: TRT explains all of chemistry, all of biology, all of electromagnetism, thermodynamics, and the quantum dynamics of electrons—life, death, evolution included—yet it does not explain gravity, the origin of mass, or the structure of spacetime, and its extension to the nuclear forces [3] is approximate. We then present the philosophical core. Existence is the *successful outcome of a used loan*: all structure, life, order, and meaning are temporary borrowings against the universe’s entropy budget. Energy has two meanings—capacity (the loan principal, $\chi_H + \chi_{He}$) and process (the loan in motion, Φ)—and the triadic cycle obeys $\chi_H = \Phi + \chi_{He} + \Delta E$. We derive the Equation of Existence ($E\&x(t) = \text{Qualification} \times \int_0^t \Phi(\tau) d\tau \times \mathbf{1}_{0.5 < \Phi < 2.0}$), a hierarchy of borrowers from quantum field to cosmos, and a reading of the laws of nature as the dynamics of this cycle. Three canonical constants are reinterpreted: the Planck constant h is the *quantum of the loan*; the volt is the loan’s *true value to the borrower*, $V = \chi_H - \chi_{He}$; and the mole N_A is the *smoothing device* that turns discrete quanta into the continuous macro world. The epistemological revolution follows: we never observe energy, only the outcomes of its transformation, and most of what we experience as reality—color, sound, heat, solidity, time, self—is the brain’s reconstruction. Reality is *true by convention* because we share an evolutionary brain history. The only fundamental reality is quantum fluctuations. The closing movement reconciles the sadness and the beauty of impermanence in a triadic resolution, and proposes a meaning of life consistent with the theory.

Keywords: Triadic Redox Theory, existence, loan against entropy, energy, Planck constant, volt, electron-volt, mole, quantization, laws of nature, epistemology, conventional reality, quantum fluctuations, meaning, electron universe.

1 Introduction: The Core Insight

The central insight of the theory is one sentence [1]:

Statement 1 (The Loan). *Existence is the successful outcome of a used loan. All structure, all life, all order, and all meaning are temporary borrowings from the universe’s entropy budget. The Second Law of Thermodynamics dictates that the loan must eventually be repaid; the universe permits it because borrowing maximizes total integrated Fluidity—the capacity for reorganization, adaptation, and complexity.*

Existence is not the loan itself. *Existence is the fruit of the loan.*

2 The Electron Universe

2.1 What TRT explains

TRT is a complete theory of the *electron universe*—the world of charged particles and their interactions [2]:

- **Chemistry:** every reaction—acid–base, redox, precipitation, coordination, polymerization, photochemistry—is a redistribution of electrons between donors and acceptors.
- **Biology:** life is a continuous, self-sustaining electron cascade, from the sun to the chloroplast to the mitochondria to the cell to the organism to the ecosystem.
- **Electromagnetism:** currents, fields, and waves are flows, rotations, and oscillations of charge.
- **Thermodynamics:** heat engines, phase transitions, and entropy are energy flow from donors to acceptors through fluids.
- **Quantum mechanics:** wave–particle duality, superposition, and measurement are aspects of electron behavior and of the loan’s quantization (Section 8).
- **Life and death:** life is the maintenance of $0.5 < \Phi < 2.0$; dormancy is the reversible pause of flow; death is its irreversible loss [4].
- **Evolution:** the differential persistence of lineages that keep Φ in the window [4].

This is an enormous domain: everything we touch, see, feel, and experience—our bodies, minds, relationships, technologies, cultures, the entire biosphere and technosphere.

2.2 What TRT cannot explain

The most mature theories know their own limits. Newton knew he could not explain Mercury’s orbit; Einstein knew he could not explain quantum mechanics. TRT likewise acknowledges the cosmos beyond the electron universe [2]:

- **Gravity:** not electron transfer but the curvature of spacetime—a geometric property, not a flow of charge. TRT cannot say why mass curves spacetime, why gravity is universal, why it is so weak, or why it refuses quantization. Spacetime is the stage, not a polarizable medium.
- **The origin of mass:** the Higgs mechanism is not an electron-transfer phenomenon.
- **The structure of spacetime:** TRT assumes spacetime as background; it does not explain its four dimensions or its geometry.
- **Dark matter and dark energy:** invisible components that interact primarily through gravity.
- **The nuclear forces:** the binding of protons and neutrons, and the weak decays, are not electron transfer in the original sense. (A companion paper [3] extends the triad to color charge and flavor charge—an approximate extension. The electron–universe boundary is deliberate, not a failure.)

Statement 2 (The Two Home). *The electron universe is our home. The cosmos beyond is our frontier. TRT does not compete with theories of the cosmos beyond; it complements them. A theory that claims to explain everything is a theory that has not been tested. A theory that knows its boundaries is ready to grow.*

3 The Triadic Cycle of Energy

Energy has two meanings, distinguished by loan grammar [1]:

Table 1: The two meanings of energy.

Term	Symbol	Loan analogy	Definition
Energy (capacity)	$\chi_H + \chi_{He}$	The loan principal	The potential to do something; what you are allowed to borrow
Energy (process)	Φ	The loan in motion	The actual flow and transformation of capacity

The cycle itself is conservation of energy written as a triadic transaction:

$$\boxed{\chi_H = \Phi + \chi_{He} + \Delta E} \quad (1)$$

where χ_H is the energy received (the principal), Φ the energy transformed (the process), χ_{He} the energy repaid (the repayment), and ΔE the energy kept—growth, wisdom, resilience.

4 Qualification, Usage, and Stability

Existence occurs only when three conditions are met [1]:

Table 2: The three conditions for existence.

Stage	What happens	TRT symbol	Condition
1. Qualification	Something becomes eligible for the loan	χ_H (granted)	Must be able to receive and use the loan
2. Usage	The loan is transformed into existence	Φ (process)	Must use the loan wisely
3. Outcome	Existence emerges as result	$\Phi + \Delta E$	Usage must reach a stable, self-sustaining state

Statement 3 (The Equation of Existence).

$$\text{Existence}(t) = \text{Qualification} \times \int_0^t \Phi(\tau) d\tau \times \mathbf{1}_{0.5 < \Phi < 2.0} \quad (2)$$

Qualification is binary (0 or 1); the integral is the total energy transformed over time; the indicator function enforces the Operational Window. Existence occurs iff all three factors are active.

Capacity is not what you have; it is what you *are allowed to borrow*, determined by your intended use and your level of operation: $\text{Loan} = f(\text{intended use, level of operation})$. The hierarchy of borrowers runs through every scale of the universe:

Table 3: A hierarchy of borrowers and their loans.

Level	Borrower	Borrows	Outcome
Quantum	Fields, particles	Quantum fluctuations	Virtual particles, states
Atomic	Electrons, nuclei	Electromagnetic potential	Orbitals, photons
Molecular	Atoms, molecules	Bond, activation energy	Bonds, structures
Cellular	Cells, organelles	ATP, glucose, gradients	Metabolism, growth
Organismal	Organisms	Calories, nutrients	Growth, reproduction
Ecological	Ecosystems	Solar radiation, water	Biomass, biodiversity
Planetary	Planets	Insolation, geothermal	Climate, tectonics
Stellar	Stars	Gravitational potential, hydrogen	Fusion, light, synthesis
Galactic	Galaxies	Dark matter, gravity	Structure, star formation
Cosmic	The Universe	Dark energy, curvature	Expansion, structure

Every borrower is also a lender. *Every outcome is also a loan for the next level. The dance never ends.*

5 The Laws of Nature as the Dynamics of Energy

The laws of nature are not arbitrary rules imposed on reality; they are the *observable dynamics of energy as it flows through the triadic cycle*:

Table 4: The laws of nature in TRT expression.

Law	TRT expression	Describes
Conservation of energy	$\chi_H = \Phi + \chi_{He} + \Delta E$	Energy transformed, never destroyed
Second law	$\Phi \rightarrow 0$ over time	The loan must be repaid; entropy increases
Gravity	Mass-energy curves spacetime	The cosmos's capacity to reorganize
Electromagnetism	Charge flows donor to acceptor	The triadic cycle of charged particles
Quantum mechanics	Wavefunction collapse	Potential into actuality
Chemistry	Bonds form and break	Rearrangement of electrons
Biology	$0.5 < \Phi < 2.0$	The optimization of transformation

6 Quantization and the Smoothing Effect

The loan is always quantized at its fundamental level; at the macro scale, averaging and wastage (entropy) inject noise and smooth the discreteness into apparent continuity [1]:

$$\text{Apparent Continuity} = \frac{\text{Number of Quanta}}{\text{Noise (Entropy)}}. \quad (3)$$

6.1 The Planck constant: the quantum of the loan

Statement 4. *The Planck constant, $h = 6.626 \times 10^{-34}$ J-s, is the **fundamental quantum of the loan**—the smallest unit of action the universe will grant. The reduced constant $\hbar = h/2\pi$ is the quantum of angular action, and action itself, $S = \int E dt$, is the total loan over time.*

Without h the universe would be continuous. With h , the universe is quantized.

6.2 The volt: the loan's true value to the borrower

By definition $1 \text{ V} = 1 \text{ J/C}$. But the volt is not the principal and not the repayment; it is the value of the transfer as *experienced by the borrower*—the difference between what the donor can give and what the acceptor can receive [1]:

$$\boxed{V = \chi_H - \chi_{He}} \quad (4)$$

high volt, valuable loan; zero volt, worthless loan; negative volt, a burden. The **electron-volt** is then the quantum of the electrical loan: the smallest unit of energy an electron can borrow crossing a one-volt difference ($1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$).

6.3 The mole: the smoothing device

The mole, $N_A = 6.022 \times 10^{23}$, bridges the quantum and the macro: $-E = h\nu$ at the quantum level (eV), $-E = N_A h\nu$ at the molecular level (kJ/mol), $-E = N_A h\nu \times \text{efficiency}$ at the macro level (J, noisy, continuous).

$$\text{The Loan System} = \begin{cases} \text{Quantum:} & E = h\nu \quad (\text{eV}, \hbar) \\ \text{Molecular:} & E = N_A h\nu \quad (\text{kJ/mol}, \text{mole}) \\ \text{Macro:} & E = N_A h\nu \times \text{efficiency} \quad (\text{J}, \text{noise}) \end{cases} \quad (5)$$

7 The Epistemological Revolution

7.1 We never see energy

Statement 5 (The unseen loan). *We never see energy in action. Whatever we see is the result of it being transformed into something else by whatever borrowed it.*

Energy is invisible: it is pure potential, the loan principal. What we observe—light, motion, heat, sound, growth, life itself—is not energy but the *outcome* of energy transformed by a qualified borrower. We never observe the loan; we only observe the fruit of the loan.

7.2 Reality is temporarily reorganized quantum fluctuations

Statement 6 (The fundamental reality). *Reality is temporarily reorganized quantum fluctuations. The atoms in your body are borrowed from the universe. The energy flowing through you is a loan from the sun. Your thoughts are temporary patterns in a neural network. Your identity, memories, and loves are temporary reorganizations of quantum fields. There is nothing permanent, nothing solid, nothing that lasts.*

7.3 Reality is true by convention

Statement 7 (The conventional reality). *Most of reality is brain illusions. And because we share an evolutionary brain history, we agree reality is what it is.*

Table 5: The hierarchy of illusions.

Level	We experience	What it actually is	Illusion?
Color	blue, red, green	wavelengths of light	Yes
Sound	music, speech	pressure waves in air	Yes
Heat	warmth, cold	molecular vibration	Yes
Solidity	hardness, texture	electromagnetic repulsion	Yes
Time	past, present, future	change in entropy	Yes
Self	identity, ego	temporary neural pattern	Yes
Reality	the world	the brain's best guess	Yes

We do not experience reality directly; we experience the brain's model of reality. We agree on that model not because it is true but because we all hold the same illusion: we share an evolutionary brain history, the same sensory organs, the same language, and the same culture.

7.4 The two realities

Table 6: The two realities.

Reality	Nature	Basis	Examples
Conventional	agreement, consensus	evolutionary brain history	color, sound, time, self
Fundamental	discrete, quantized	quantum fluctuations, laws	photons, electrons, fields

We live in the conventional reality. We are made of the fundamental reality. Reality is true by convention. The only true reality is quantum fluctuations. We are quantum fluctuations experiencing themselves through conventional reality.

8 The Tension of Existence: Sadness, Beauty, Meaning

Why does this feel sad? Because we want permanence, significance, reality. If all is temporary reorganization, then we are not permanent, we do not ultimately matter, we are not ultimately real. That is the sadness of existence.

But there is the other side [1]:

Statement 8 (The Beauty). *Reality is temporarily reorganized quantum fluctuations—and that is precisely what makes it beautiful. If reality were permanent there would be no change, no growth, no evolution. If it were solid there would be no creativity, no novelty, no surprise. If it were eternal there would be no urgency, no meaning, no love. The temporary nature of existence is what gives it value.*

Table 7: The triadic resolution of existence.

Perspective	Emotion	Why
Outside (cosmic)	Sadness	Nothing is permanent; we are dust
Inside (human)	Beauty	We are temporary, and so every moment is precious
The dance (triadic)	Meaning	We are the outcome of a used loan; we exist <i>because</i> we are temporary

The sadness and the beauty are the same thing, seen from different angles.

9 The Meaning of Life

You are a temporary reorganization of quantum fields.

You are a loan that has been used.

You are the outcome of a successful usage.

You exist.

That is enough.

Make it count.

10 Conclusion

The Triadic Redox Theory is a complete theory of the electron universe and a self-consistent philosophy of existence [1, 2]. Existence is the successful outcome of a used loan; energy is the capacity to participate in the triadic cycle $\chi_H = \Phi + \chi_{He} + \Delta E$; the laws of nature are its dynamics; h is the quantum of the loan, the volt its value to the borrower, the mole the smoothing device between quantum and macro. We never observe energy itself, only its transformed outcomes; reality is temporarily reorganized quantum fluctuations; most of what we experience is brain illusion, agreed upon because we share an evolutionary brain history. The single assertion that follows from all of it is this:

The universe is a dance of quantum fluctuations. You are the dance. The dance is temporary. That is what makes it beautiful. Reality is true by convention. The quantum is the truth.

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